Midterm Exam II — Part 2

PPEM 454 — Viral Ecology

Name:

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Date:

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Workflow: Incidence matrix, alpha diversity, beta diversity, ordination, and viral community composition.

Before knitting: Install `ViralExamR`, `vegan`, `ggplot2`, and `tidyr` in the R console. The seed can be changed in the YAML header.

1. Load packages and choose a seed

```
library(ViralExamR)
library(vegan)
library(ggplot2)
library(tidyr)
seed <- params$seed
```

Q1. Write down your seed number (at least 4 digits).

---

3 8 4 5

2. Create the virus incidence matrix

```
inc_matrix <- create_incidence_matrix(10, 8, seed = seed)
inc_matrix
```

```
##         sampleA sampleB sampleC sampleD sampleE sampleF sampleG sampleH
## virus1       38      74      25      27      39      49       0       0
## virus2       24       0      32       0       0       0      64      20
## virus3      106      69       0       7      44       0       0      53
## virus4        1      55      56      88      57     125      94      14
## virus5      142     118      77       0       6      63     139      41
## virus6       80       0      72      65      59       0       0     126
## virus7       52       1       0      95     146       0       0     106
## virus8       72     176      89       0       0     110      35       0
## virus9      104       0       6       0      80      33      73      40
## virus10      64      75      79       3       0      35      86      14
```

```
# 3. Calculate the total abundance for each virus
rowSums(inc_matrix)
```

```
##  virus1  virus2  virus3  virus4  virus5  virus6  virus7  virus8  virus9 virus10
##     252     140     279     490     586     402     400     482     336     356
```

```
community <- t(inc_matrix)  # vegan: samples in rows, taxa in columns
```

Q2. What does total abundance by sample represent?

3. Alpha diversity

```
Richness <- specnumber(community)
Richness
```

```
## sampleA sampleB sampleC sampleD sampleE sampleF sampleG sampleH
##      10       7       8       6       7       6       6       8
```

Q3. What is richness for each sample? Which has highest and lowest richness? What does this suggest?

```
Shannon <- diversity(community, index = "shannon")
Shannon
```

```
##  sampleA  sampleB  sampleC  sampleD  sampleE  sampleF  sampleG  sampleH
## 2.096501 1.715686 1.915684 1.428405 1.728913 1.661699 1.716109 1.804268
```

Q4. What is the Shannon index for each sample? Which samples have the highest and lowest diversity? Interpret.

```
Simpson <-  diversity(community, index = "simpson")
Simpson
```

```
##   sampleA   sampleB   sampleC   sampleD   sampleE   sampleF   sampleG   sampleH
## 0.8657610 0.8022838 0.8416800 0.7318436 0.7957645 0.7885963 0.8083507 0.8016640
```

Q5. What is the Simpson diversity index for each sample? Which have the highest and lowest diversity? Interpret.

```r
Inverse_Simpson <-  diversity(community, index = "invsimpson")
Inverse_Simpson
```

```
##  sampleA  sampleB  sampleC  sampleD  sampleE  sampleF  sampleG  sampleH
## 7.449402 5.057754 6.316321 3.729168 4.896307 4.730286 5.217865 5.041949
```

Q6. What is the inverse Simpson index for each sample? Which have the highest and lowest diversity? Interpret.

```r
Evenness <- ifelse(Richness > 1, Shannon / log(Richness), NA_real_)
Evenness
```

```
##   sampleA   sampleB   sampleC   sampleD   sampleE   sampleF   sampleG   sampleH
## 0.9104990 0.8816884 0.9212494 0.7972081 0.8884855 0.9274116 0.9577785 0.8676697
```

Q6 (Evenness). What is Pielou's evenness for each sample? Which has highest and lowest evenness? Interpret.

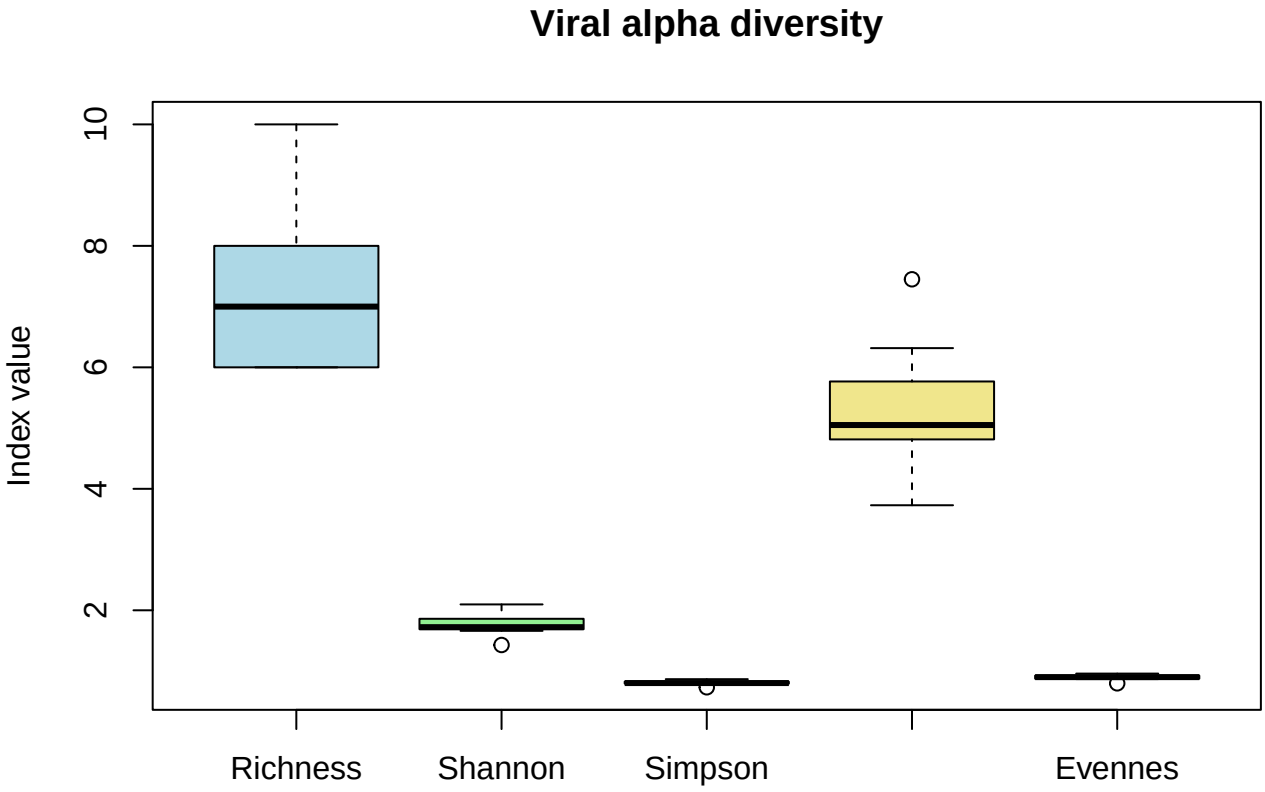
---

4. Summary and alpha-diversity plots

```r
Alpha_diversity_results <- data.frame(
  Sample = rownames(community),
  Abundance = rowSums(community),
  # vegan package using diversity() function
  Richness = Richness,
  Shannon = Shannon,
  Simpson = Simpson,
  Inverser_Simpson = Inverse_Simpson,
  Evenness = Evenness,
  row.names = NULL
)
print(Alpha_diversity_results)
```

```
##    Sample Abundance Richness  Shannon   Simpson Inverser_Simpson  Evenness
## 1 sampleA       683       10 2.096501 0.8657610         7.449402 0.9104990
## 2 sampleB       568        7 1.715686 0.8022838         5.057754 0.8816884
## 3 sampleC       436        8 1.915684 0.8416800         6.316321 0.9212494
## 4 sampleD       285        6 1.428405 0.7318436         3.729168 0.7972081
## 5 sampleE       431        7 1.728913 0.7957645         4.896307 0.8884855
## 6 sampleF       415        6 1.661699 0.7885963         4.730286 0.9274116
## 7 sampleG       491        6 1.716109 0.8083507         5.217865 0.9577785
## 8 sampleH       414        8 1.804268 0.8016640         5.041949 0.8676697
```

```r
boxplot(
  Richness,
  Shannon,
  Simpson,
  Inverse_Simpson,
  Evenness,
  names = c("Richness", "Shannon", "Simpson", "Inverse Simpson", "Evennes"),
  ylab = "Index value",
  main = "Viral alpha diversity",
  col = c("lightblue", "lightgreen", "salmon", "khaki", "gray")
)
```



Q7. Write a figure caption suitable for a scientific paper describing the alpha-diversity figure.

5. Beta diversity: Bray–Curtis

```r
bray <- vegdist(community, method = "bray")
bray_matrix <- as.matrix(bray)

# Remove self-comparisons (distance to itself = 0)
diag(bray_matrix) <- NA
bray_matrix
```

```
##           sampleA   sampleB   sampleC   sampleD   sampleE   sampleF   sampleG
## sampleA        NA 0.4196643 0.3905273 0.6797521 0.4973070 0.5591985 0.4275980
## sampleB 0.4196643        NA 0.3605578 0.7819461 0.7097097 0.3652085 0.4655335
## sampleC 0.3905273 0.3605578        NA 0.5866852 0.6493656 0.3560517 0.3851133
## sampleD 0.6797521 0.7819461 0.5866852        NA 0.3156425 0.6628571 0.7654639
## sampleE 0.4973070 0.7097097 0.6493656 0.3156425        NA 0.6808511 0.7049892
## sampleF 0.5591985 0.3652085 0.3560517 0.6628571 0.6808511        NA 0.4260486
## sampleG 0.4275980 0.4655335 0.3851133 0.7654639 0.7049892 0.4260486        NA
## sampleH 0.4512306 0.7494908 0.6070588 0.4735336 0.3633136 0.7539204 0.7149171
##           sampleH
## sampleA 0.4512306
## sampleB 0.7494908
## sampleC 0.6070588
## sampleD 0.4735336
## sampleE 0.3633136
## sampleF 0.7539204
## sampleG 0.7149171
## sampleH        NA
```

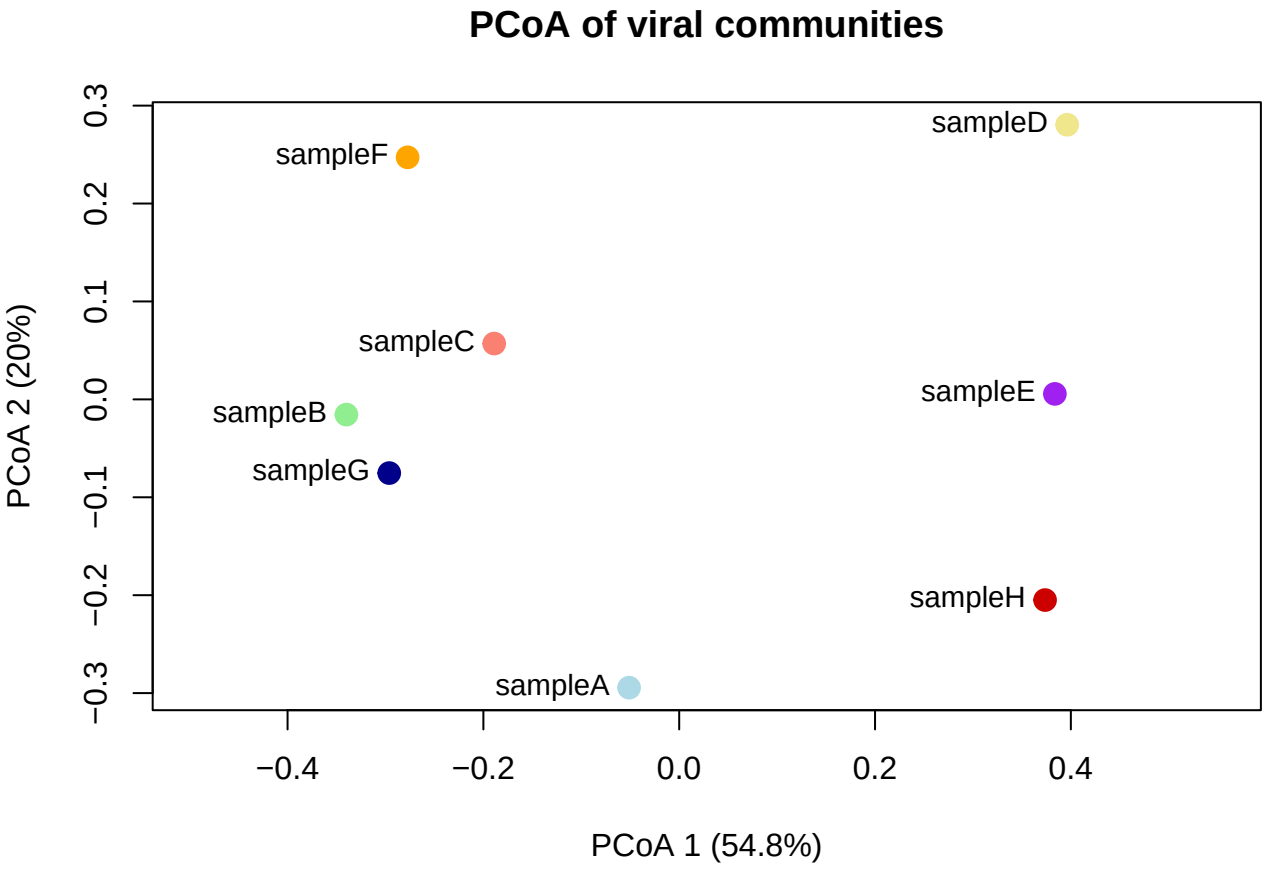
Q8. What is Bray–Curtis dissimilarity? Which sample pairs have the highest and lowest dissimilarity? Interpret.

6. Principal Coordinates Analysis (PCoA)

```r
pcoa <- cmdscale(bray, k = 2, eig = TRUE, add = TRUE)

# Percentage of variation represented by each axis
percent <- 100 * pcoa$eig / sum(pcoa$eig[pcoa$eig > 0])

# Base R plot
{plot(pcoa$points,  pch = 19,
     col = c("lightblue","lightgreen", "salmon", "khaki", "purple",
             "orange", "blue4", "red3"),
     cex = 1.5, asp = 1,
     xlab = paste0("PCoA 1 (", round(percent[1], 1), "%)"),
     ylab = paste0("PCoA 2 (", round(percent[2], 1), "%)"),
     main = "PCoA of viral communities")
text(pcoa$points, labels = rownames(community), pos = 2,  cex = 0.9)
}
```

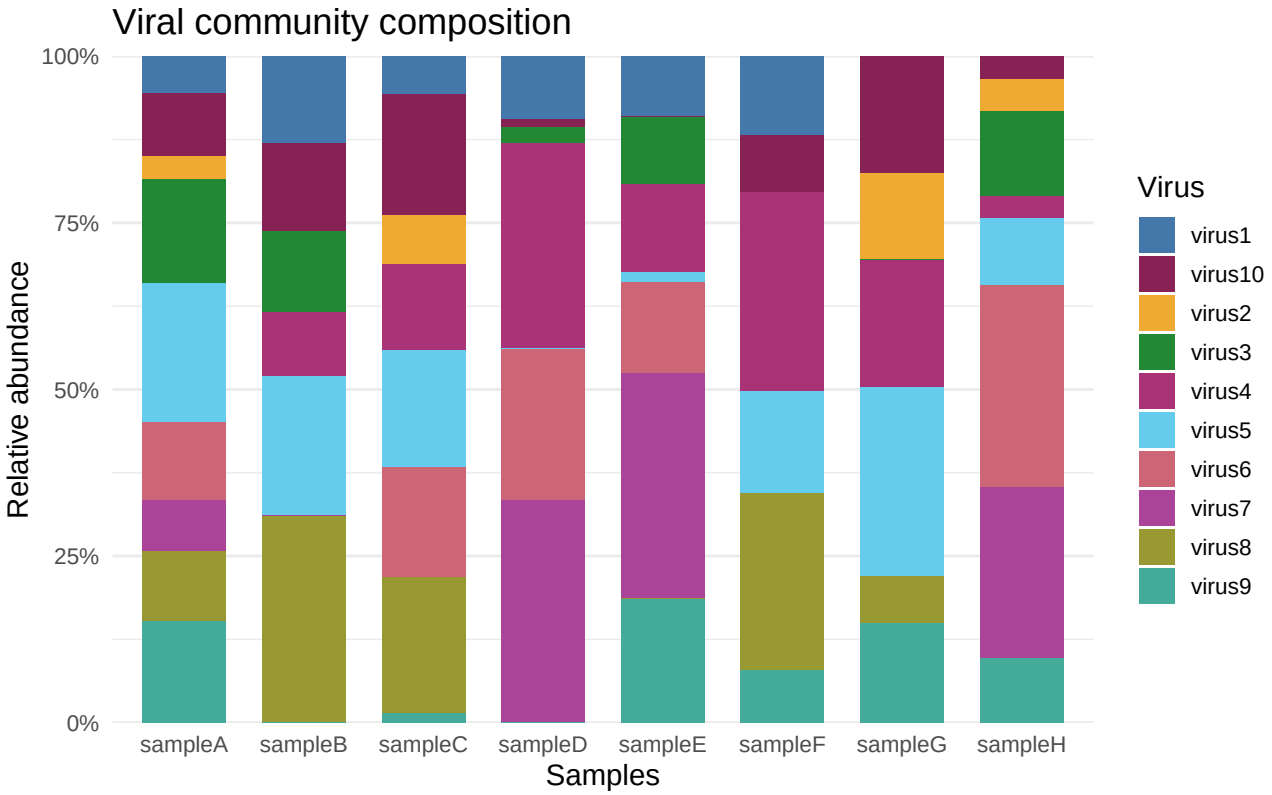


Q9. Describe the ordination patterns. Which samples cluster together, and which appear most different?

7. Viral community composition

```
relative_abundance <- decostand(community, method = "total")
# format data matrix
composition <- data.frame(Sample = rownames(relative_abundance),
                          relative_abundance, check.names = FALSE )
# transform data matrix to a long table to be compatible with ggplot
composition_long <- pivot_longer(composition, cols = -Sample,
                                 names_to = "Virus", values_to = "Proportion")
# plot the community compositio
ggplot(composition_long, aes(x = Sample, y = Proportion, fill = Virus))+
  geom_col(width = 0.7) +
  scale_y_continuous(labels = function(x) paste0(round(100 * x), "%"),
                     expand = expansion(mult = c(0, 0))) +
  scale_fill_manual(values = c("virus1" = "#4477AA", "virus2" = "#EEAA33",
                               "virus3" = "#228833","virus4" = "#AA3377",
                               "virus5"  = "#66CCEE", "virus6"  = "#CC6677",
                               "virus7"  = "#AA4499", "virus8"  = "#999933",
```

```
                              "virus9"  = "#44AA99", "virus10" = "#882255"))+
labs(title="Viral community composition", x="Samples", y="Relative abundance",
     fill="Virus")+
theme_minimal(base_size = 13) +
theme(panel.grid.major.x = element_blank())
```



Q10. Describe the community-composition patterns and how they relate to the ordination.